

Professional Summary

AI/LLM Engineer with hands-on experience building and deploying production-grade GenAI systems, agentic AI frameworks, and LLM-powered applications. Core strengths include RAG pipelines, multi-agent orchestration, LLM fine-tuning (QLoRA/LoRA), prompt and context engineering, MCP tooling, and evaluation/observability. Strong Python foundation with end-to-end deployment experience using Docker, CI/CD, and cloud platforms.

Technical Skills

AI / LLM

LLM Integration, RAG Pipelines, Agentic AI Frameworks, Prompt Engineering, Context Engineering, Multi-Agent Systems, Tool-Using Agents, Fine-tuning (QLoRA/LoRA), MCP / Tool Protocols, LangChain / LlamaIndex, Vector Databases, Evaluation & Observability, AI Governance, Local LLM Deployment, NLP, Ollama

Languages

Python, TypeScript, JavaScript, Rust, C++, C, PHP, HTML/CSS

Frontend

React, React Native, Tailwind CSS, Bootstrap, Monaco Editor, Next.js

Backend

Node.js, Flask, Express, FastAPI, Axum (Rust), REST APIs, WebSockets, Laravel, Supabase

Databases

MySQL, MongoDB, Redis, SQLite

DevOps

Docker, GitHub Actions, Nginx, Gunicorn, Prometheus, Linux

Vision / ML

YOLO, OpenCV, Computer Vision

Projects

NexusAOS AI/AGENTIC

Governance-first agentic OS with multi-agent swarm orchestration, LLM fine-tuning, and 30+ MCP tools.

- Built multi-agent swarm executor with collision detection, namespace isolation, quorum voting, and atomic fission in Python + asyncio
- Integrated local LLM inference with Phi-4-Mini QLoRA adapters, AB/AP balance enforcement, and constitution-guided routing
- Exposed 30+ MCP tools via FastMCP for metabolic, planning, immune, and physical substrate control
- Designed adapter-routed inference with curriculum learning, evaluation harness, and model benchmarking

[Source Code](#) [Documentation](#)

nexus-kernel AI/AGENTIC

Production-ready Rust microkernel for local-first AI with event-sourced governance, OpenAI/Anthropic streaming, and 981 tests.

- Built Rust 2024 microkernel with 12 workspace crates, 981 passing tests, 0 clippy warnings, and full CI/CD
- Implemented event-sourced append-only audit trail, policy engine with trust tiers, and provider-swappable model interface
- Integrated OpenAI-compatible and Anthropic streaming with real-time token streaming into TUI/GUI
- Delivered native terminal emulation (PTY + VT100 + Zig parser), SSH multiplexing, and multi-interface CLI/TUI/GUI/RPC

[Source Code](#) [Documentation](#)

SeshaOS AI/AGENTIC

NexusAOS v2 — governance-first, local-first AI OS with specialist local models and LiteLLM-compatible routing.

- Architected specialist model stack: Gemma 4 12B (Planner), Qwen3-Coder 30B (Implementation), Qwen3.5 9B (Vision)
- Designed kernel-centric governance where models propose actions and the kernel validates/records every state change
- Implemented LiteLLM-compatible proxy support for NVIDIA NIM and other model providers

- Maintained event-sourced architecture with reversible, permissioned actions and offline-first execution

[Source Code](#) [Documentation](#)

Vyākṛti FLAGSHIP

Sanskrit-oriented programming language with complete compiler pipeline and browser-based IDE. 123 tests.

- Complete compiler pipeline: lexer → parser → type checker → bytecode compiler — all built from scratch in Rust
- Browser-based IDE with React, Monaco Editor, syntax highlighting, autocomplete, and diagnostics
- Rust (axum) backend with compile, REPL, LSP, and file management endpoints via REST + WebSocket
- 123 tests covering the full pipeline, including a self-hosting corpus

[Source Code](#) [Documentation](#)

AIM — Attendance Information Manager PRODUCTION-READY

Production-grade Flask + MySQL platform with Argon2id auth, Prometheus monitoring, and CI/CD.

- Production-grade Flask + MySQL platform with Argon2id auth, CSRF protection, brute-force lockout, breached-password scanning, JWT sessions, and strict CSP/HSTS headers
- Deployed Prometheus metrics, structured JSON logging, Chart.js analytics, FullCalendar scheduling, Docker Compose, and GitHub Actions CI/CD; 101 automated pytest tests

[Source Code](#)

FWRS — Food Waste Reduction System PRODUCTION-READY

Food waste optimization platform using linear programming and route optimization.

- Built linear programming models to minimize food waste and optimize distribution routes
- Implemented route optimization algorithms for efficient collection and delivery scheduling
- Integrated Folium-based mapping and visualization for route planning and monitoring
- Deployed as a full-stack platform with real-time tracking and analytics dashboard

[Source Code](#)

Experience

Industry 5.0 Industrial Automation Trainee

CodenPlay Robotics

VIT Vellore / Online

- Built and debugged PLC automation logic (ladder logic, structured text) in CODESYS
- Simulated industrial process workflows in Factory I/O
- Implemented flow-based control/integration in Node-RED
- Covered HMI design, industrial communication, process monitoring, and the debugging→testing→deployment workflow

Aug – Nov 2025

Education

VIT Vellore

B.Tech in Computer Science and Engineering

Graduated 2026